

**BEFORE YOU READ — ANSWER FROM MEMORY**

1. Which three categories of liver tests reflect hepatocyte integrity, biliary excretion, and synthetic function respectively?
2. What ammonia-related mechanism explains hepatic encephalopathy, and what mnemonic covers its precipitants?
3. What are the two defining pathologic features of cirrhosis, on gross and microscopic exam?

Guessing wrong before reading still improves retention — attempt all three, then check as you go.

1 Laboratory assessment of liver function

Liver integrity is assessed through three elements of liver structure/function.

Element	Tests	Elevated when...
1. Hepatocyte integrity	Cytosolic enzymes: AST, ALT, LDH	Hepatocyte destruction (e.g. by viruses)
2. Biliary excretory function	Serum bilirubin (total = unconjugated + conjugated; direct = conjugated) · urine bilirubin · serum bile acids · plasma membrane enzymes (alkaline phosphatase, GGT)	Destruction/ obstruction of the biliary passage
3. Hepatocyte synthetic function	Proteins: serum albumin , coagulation factors (PT, PTT) · metabolism: serum ammonia , aminopyrine breath test	Impaired synthetic/metabolic capacity

2 Liver failure — the frame

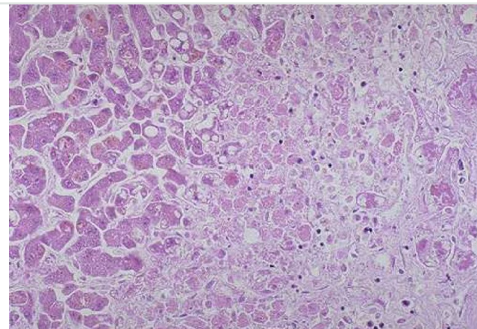
What	Inability of the liver to perform its normal synthetic, metabolic and excretory functions.
Three types	Acute liver failure · chronic liver failure/cirrhosis · acute-on-chronic liver failure (a sudden acute insult superimposed on years of stable, compensated chronic liver disease).

Acute liver failure

Definition	Acute liver illness producing hepatic encephalopathy within 6 months of initial diagnosis.
Causes	Acetaminophen (~50% of US cases) · hepatitis B/E (predominate in Asia) · acute hepatitis A · autoimmune hepatitis · other drugs/toxins.
Gross pathology	Liver small and shrunken (loss of parenchyma), bile-stained, soft, congested.
Microscopic	Large zones of destruction around islands of regenerating hepatocytes; scarring mostly absent (acute process).
Clinical features	Nausea, vomiting, jaundice, fatigue → encephalopathy, coagulation defects, portal hypertension with ascites .



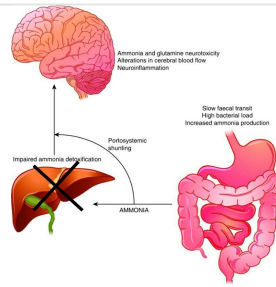
Massive liver necrosis: small, bile-stained, soft, congested liver



Acetaminophen overdose: extensive hepatocyte necrosis (right) adjacent to dying hepatocytes (left)

Hepatic encephalopathy

What	Symptoms ranging from subtle behavioral change to confusion, stupor, coma, death. Typical neurologic sign: asterixis .
Mechanism	Believed caused by elevated ammonia , correlating with impaired neuronal function and cerebral edema (also: glutamine neurotoxicity, altered cerebral blood flow, neuroinflammation).



Hepatic encephalopathy pathway: impaired hepatic ammonia detoxification + portosystemic shunting + gut ammonia production → brain

Precipitants of hepatic encephalopathy — mnemonic “HEPATICS”: Hemorrhage in GIT · Excess protein in diet · Potassium depletion (diuretics) · Alkalosis, azotemia · Trauma · Infections (bacterial, peritonitis, etc.) · Constipation (colon surgery) · Sedatives.

Chronic liver failure and cirrhosis

What	Cirrhosis = diffuse transformation of the liver into regenerative parenchymal nodules surrounded by fibrous bands. Not a specific entity — a morphologic change, often from severe chronic liver disease.
Causes	Chronic hepatitis B · chronic hepatitis C · NAFLD · alcohol-related liver disease.
Key caveats	Prognosis is variable. Not all chronic liver disease ends in cirrhosis; not all cirrhosis leads to end-stage liver disease.

Clinical features and consequences

Feature	Detail
Asymptomatic	~ 40% of individuals
Nonspecific	Anorexia, weight loss, weakness
Liver failure signs	Jaundice, encephalopathy, coagulopathy; chronic severe jaundice → pruritus
Portal hypertension	From increased vascular resistance + increased portal blood flow
Portosystemic shunts	Dilated collaterals where portal/systemic circulations share capillary beds — esophagogastric varices, caput medusae, rectal varices/hemorrhoids
Ascites	Transudate type
Splenomegaly	With hypersplenism
Hyperestrogenemia	Angiomas, gynecomastia, and others
Malignancy risk	Predisposes to hepatocellular carcinoma

Pathologic features

Gross	Microscopic
Multiple regenerative nodules throughout the liver, separated by bridging fibrosis	Diffuse disruption of architecture (loss of normal central-portal relationship); bridging fibrous septa; parenchymal nodules of regenerating hepatocytes



Cirrhosis, microscopic: regenerating nodules (circled) separated by bridging fibrous septa (lines)



Cirrhosis, gross: multiple regenerative nodules throughout the liver

P Must-remember pearls

- Three pillars of liver testing: **hepatocyte integrity (AST/ALT/LDH)** · **biliary excretory function (bilirubin, ALP, GGT)** · **synthetic function (albumin, PT/PTT, ammonia)**.
- **Acute liver failure** = encephalopathy within **6 months** of diagnosis; **acetaminophen** is the top US cause, **HBV/HEV** top causes in Asia. Liver is small/shrunken with minimal scarring (acute).
- Hepatic encephalopathy is driven by **elevated ammonia** from impaired hepatic clearance and portosystemic shunting; remember precipitants with **HEPATICS**.
- **Cirrhosis = regenerative nodules + bridging fibrosis**, not a specific disease; **not all chronic liver disease → cirrhosis, and not all cirrhosis → end-stage disease**.
- Cirrhosis complications flow from **portal hypertension** (varices, caput medusae, ascites, splenomegaly) and **loss of synthetic/detox function** (coagulopathy, encephalopathy, hyperestrogenemia); cirrhosis predisposes to **HCC**.

Q Test yourself

1. A patient with acute viral hepatitis has markedly elevated AST and ALT. These enzymes reflect which aspect of liver function?

- A. Hepatocyte integrity
- B. Biliary excretory function
- C. Hepatocyte synthetic function
- D. Portal perfusion
- E. Hepatic clearance of ammonia

2. A patient ingests a large overdose of acetaminophen and develops hepatic encephalopathy within one week. This is best classified as:

- A. Chronic liver failure
- B. Acute-on-chronic liver failure
- C. Cirrhosis
- D. Acute liver failure
- E. Hepatorenal syndrome

3. A cirrhotic patient becomes constipated after surgery and develops worsening confusion and asterix. Which mechanism best explains this precipitant of hepatic encephalopathy?

- A. Direct hepatotoxicity of surgical anesthesia
- B. Acute viral superinfection
- C. Increased ammonia production and absorption from stasis in the colon
- D. Loss of portal vein patency
- E. Esophageal variceal bleeding

4. Liver biopsy in a patient with long-standing hepatitis C shows regenerative nodules of hepatocytes separated by bridging fibrous septa, with loss of normal central-portal architecture. This is:

- A. Acute liver failure
- B. Hepatocellular adenoma
- C. Hepatic hemangioma
- D. Acute hepatitis
- E. Cirrhosis

5. A cirrhotic patient develops dilated collateral veins at sites where portal and systemic circulations share capillary beds, including esophageal varices and caput medusae. This is a direct consequence of:

- A.** Hepatic encephalopathy
- B.** Portal hypertension
- C.** Hyperestrogenemia
- D.** Coagulopathy
- E.** Hypersplenism

ANSWERS — COVER UNTIL YOU'VE COMMITTED

- 1 • A** — AST, ALT, and LDH are cytosolic hepatocyte enzymes that elevate with hepatocyte destruction, reflecting hepatocyte integrity.
- 2 • D** — Acute liver failure is defined as acute liver illness producing hepatic encephalopathy within 6 months of diagnosis; acetaminophen is the leading US cause.
- 3 • C** — Constipation (e.g. after colon surgery) is a recognized precipitant of hepatic encephalopathy via increased ammonia production/absorption, as in the HEPATICS mnemonic.
- 4 • E** — Cirrhosis is defined by diffuse transformation into regenerative parenchymal nodules surrounded by fibrous bands, with loss of the normal central-portal relationship.
- 5 • B** — Sustained portal hypertension leads to portosystemic shunting and dilation of collateral vessels, producing esophagogastric varices, caput medusae, and rectal varices.

SPACED REVIEW

☐ Day 1

☐ Day 3

☐ Day 7

☐ Day 21

☐ Pre-exam

Pathology GIT Block · MEDucation by Mattin Majid · summarised from SGL5 (titled “Liver, pancreas and biliary system” — slide content covers liver only), Dr Savan Saeed Azeez, Hawler Medical University
Study alongside the lecture, not instead of it.